

Semantic categorization of DDI metadata

Abstract

The work presents a modelling and implementation effort in semantic categorization of DDI metadata from the UK Data Archive (<http://www.data-archive.ac.uk/>), the UK ESRC Research Catalogue (<http://www.esrc.ac.uk/impacts-and-findings/research-catalogue/index.aspx>), and GESIS – Leibniz Institute for the Social Sciences (<http://www.gesis.org/>). The categorization output is then linked with a few common ontologies and semantic data sources that are of interest to ENGAGE project (<http://www.engage-project.eu>).

Motivation

ENGAGE project (<http://www.engage-project.eu>) aims at supplying the open data infrastructure and is largely oriented on the open data re-use by research communities. ENGAGE has identified DDI as an important framework to cater for the ENGAGE user needs, specifically for researchers in social science. One of the fundamental use case scenarios in ENGAGE is data discovery for which Linked Data is an important means, and linking to DDI metadata is a gateway to the actual data that are not necessarily in DDI format, and not necessarily in open access.

The ongoing effort of DDI Semantic Group (<http://www.gesis.org/en/events/gesis-workshops/ddi/>), as well as the due attention to the Linked Open Data agenda in DDI Users Group meetings (http://www.iza.org/conference_files/EDDI2011/call_for_papers/EDDI11%20B2%20-%20Nicole%20Quitzs%20-%20The%20Registration%20Agency,%20DDI%20and%20Linked%20Open%20Data.ppt) have raised hopes for a substantial amount of DDI data published as Linked Data in near future which would make the aforementioned task of linking DDI and open data easier for ENGAGE. Deep semantic modelling of DDI, however, may require a challenging project of its own, or a few, so the categorization (processing against a certain model) of DDI metadata and further linking them with the open data through reasonable reference points is an approach that ENGAGE (which is not a research but an infrastructure project) currently considers most realistic.

DDI metadata and experiments with them

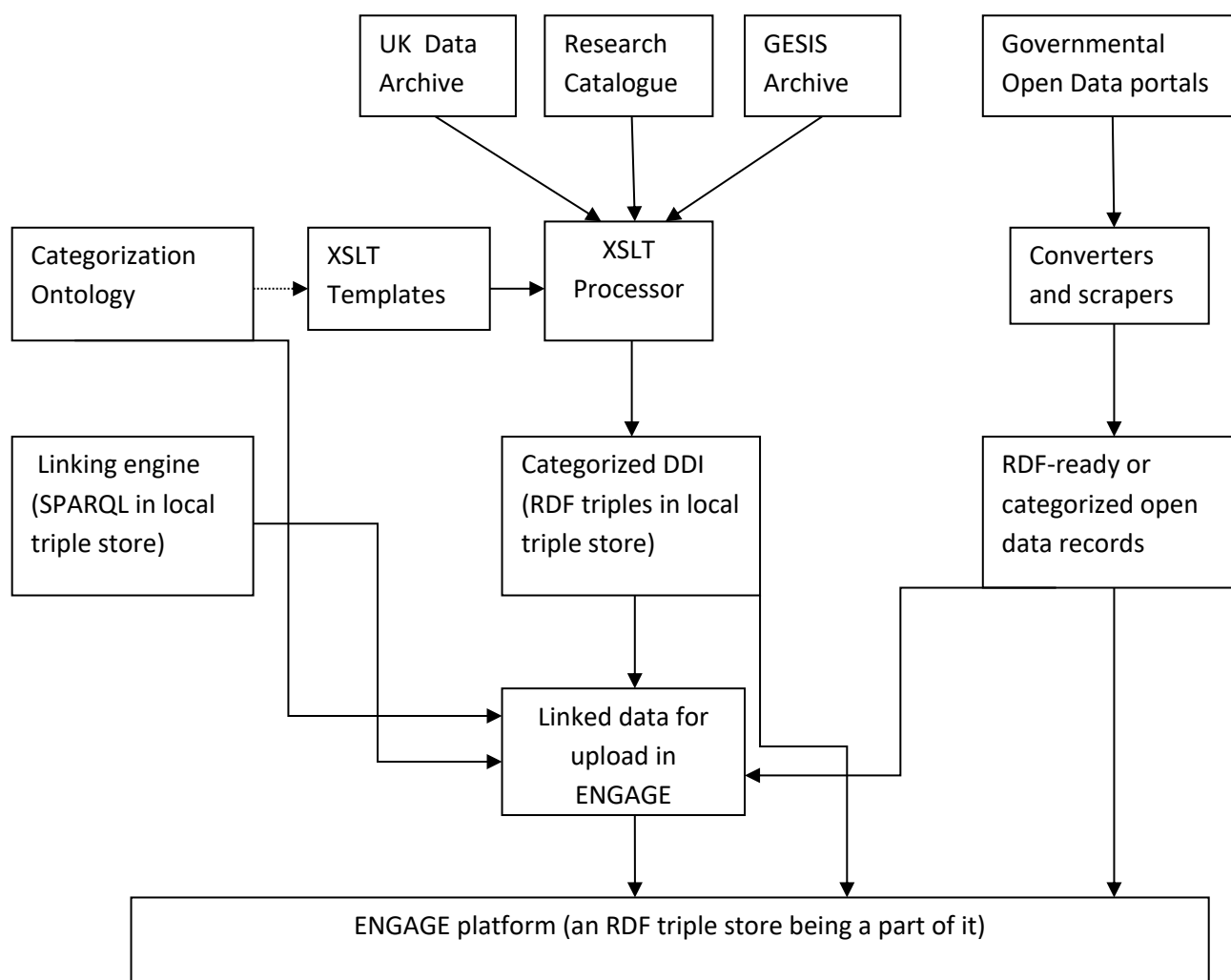
Three DDI metadata sources have been chosen for the experiments on semantic categorization: the UK Data Archive, the UK ESRC Research Catalogue, and GESIS archive

(please see references in the Abstract). Re-use of GESIS metadata is subject to regulation and may require an additional agreement with the owner so we used them for experiments only; the UK Data Archive metadata are freely available for harvesting through OAI interface so working with them may have a practical goal of linking them with other open data.

The UK ESRC Research Catalogue stores details of research funding and research outputs in social science; these details can be exposed as Dublin Core via another OAI interface. The Research Catalogue and the UK Data Archive records can be linked with each other via grant references, and we actually tried it as an interesting by-product of our DDI categorization.

We then identified the DDI tags that have a potential of linking them with a few common ontologies and semantic data sources: Dublin Core, FOAF, DBpedia, as well as with the emerging SKOS version of HASSET thesaurus (<http://hassetukda.wordpress.com/>) and the semantic representation of CERIF (<http://www.eurocris.org/Index.php?page=CERIFreleases&t=1>); we then produced RDF triples based on these tags and on a simple “DDI categorization ontology” with a few semantic inference rules in it.

We used XSLT transformations and Sesame triple store (<http://www.openrdf.org/>) with OWLIM inference engine (<http://owlim.ontotext.com>) as the toolset; the following schema represents the data and control workflow:



Results

A few hundred DDI datasets have been harvested and are under semantic categorization; the samples of categorization output will be presented in the DDI Users Group meeting.

A simple categorization ontology has been produced that includes a mapping between semantically categorized DDI records and common ontologies, as well as “sockets” for further inclusion of the specialized semantic resources. The ontology is being amended with feedback from the actual DDI records categorization; it will be also presented in the DDI Users Group meeting.

The efforts of other working groups on semantic modelling of HASSET and CERIF are being monitored; depending on the maturity of this effort by the time of the meeting, the results of mashing up the categorized DDI records with the appropriate vocabularies and information resources will be also presented.

We hope this exercise on DDI categorization may contribute to the semantic modelling and implementation effort in a wider DDI community; ENGAGE project (which is, again, an infrastructure project with only limited resource for the applied research) is open for collaboration with all interested parties.